



KG-UQ: Knowledge Graph-Based Uncertainty Quantification for Long Text in Large Language Models

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Abstract

With the commercialization of large language models (LLMs) and their integration into daily life, addressing their susceptibility to hallucinations—unfactual information in generated outputs—has become an urgent priority. Existing uncertainty quantification (UQ) methods often rely on access to LLMs’ internal states, which is unavailable for closed-source models like GPTs, or are primarily designed for short text. Current research on long text typically evaluates sentences individually, overlooking smaller semantic units that better capture the text’s complexity. Recognizing the potential of knowledge graphs (KGs) to extract structured relationships from unstructured text, we propose KG-UQ, a UQ method leveraging KGs to address the semantic intricacies of long text. Our approach involves constructing KGs from long-text outputs and utilizing their embeddings to estimate uncertainties. Through our analysis, we demonstrate that knowledge graphs are an effective tool for decomposing long text into fundamental statements. However, we also highlight the increased uncertainty introduced during KG construction, stemming from inherent challenges in accurately capturing all semantic information.

CCS Concepts

• **Computing methodologies** → **Natural language generation; Knowledge representation and reasoning.**

Keywords

Large Language Models, Hallucinations, Uncertainty Quantification, Knowledge Graphs

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1 Introduction

With the rapid development of Large Language Models (LLMs), their exceptional performance in natural language processing (NLP) tasks has been well illustrated [2, 4, 26, 37, 45], alongside their applications in various other areas [3, 32, 33, 44]. Applications, such as ChatGPT, have seamlessly integrated into everyday life. However, LLMs are often prone to hallucinations, generating responses that may be unfactual or unfaithful. Traditionally, researchers have relied on human evaluation, manually verifying decomposed atomic information. Alternatively, approaches have been developed to quantify the factuality of LLM responses or assess the confidence levels of LLMs in their outputs.

Existing uncertainty quantification (UQ) methods predominantly rely on accessing the internal states of LLMs, such as token likelihoods [8, 20, 41]. However, with the increasing commercialization of LLMs, closed-source proprietary models like GPTs restrict access to such internal states, offering only API-level interaction. Another significant challenge lies in handling the complexity of long text. While Natural Language Inference (NLI) models are often used to determine whether a piece of text is supported by its source, they perform well when dealing with fragmented information. However, long text is known for its intricate semantic structure, making it challenging to verify whether an entire passage is supported by its knowledge source. In existing literature, LUQ [42] tackle this by analyzing long text sentence by sentence, using NLI to determine whether each sentence is supported. Another approach [12] involves decomposing LLM responses into factoids and clustering these factoids by meaning to evaluate factuality.



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Knowledge graphs (KGs), widely recognized for their ability to represent structured data and organize relationships derived from unstructured data, and have demonstrated significant potential in tasks like retrieval-augmented generation. GraphEval [38] has picked up on this and leveraged this capability by feeding LLM outputs into a KG construction prompt and then evaluating the factuality of each generated triple using NLI.

Building on insights from current research, we propose multiple KG-UQ, an uncertainty quantification method based on knowledge graphs, with comparisons of existing UQ methods performance by measuring the correlation with factuality scores. The key contributions of our work are summarized as follows:

- We introduce knowledge graphs (KG) as a tool for extracting logical information, fully leveraging the capabilities of KGs to address the limitations of current uncertainty quantification (UQ) methods for long text.
- Through analysis, we uncover the inherent limitations of current knowledge graph construction methods due to their intrinsic uncertainty.
- Through extensive experiments across different models and datasets, we demonstrate the superior generalization ability of our algorithm, surpassing the state-of-the-art (SOTA) methods.

2 Related Work

2.1 Uncertainty Quantification in Machine Learning Models

Uncertainty quantification (UQ) [10, 15, 27, 39, 40], plays a crucial role in machine learning, helping models provide predictions along with measures of confidence. Traditional approaches include Bayesian methods, such as Bayesian Neural Networks (BNNs), which use posterior distributions to capture uncertainty in model parameters [27]. Non-Bayesian methods, like Monte Carlo Dropout [15] and Deep Ensembles [10], offer more practical alternatives for estimating predictive uncertainty. Before the rise of Large Language Models (LLMs), UQ was already a key area of research in machine learning [15]. Uncertainty is typically divided into two categories: aleatoric and epistemic uncertainty [7, 16]. Aleatoric uncertainty, also known as statistical uncertainty, refers to the natural randomness in data or outcomes caused by inherent variability [18]. Epistemic uncertainty, on the other hand, arises from incomplete knowledge, such as missing data or uncertainty in model parameters [18]. While aleatoric uncertainty cannot be reduced, epistemic uncertainty can often be addressed through better models or more data.

2.2 Uncertainty in Knowledge Graphs

Knowledge graphs (KGs) are widely used to represent structured data and relationships, but effectively managing uncertainty is key to their usefulness. Probabilistic knowledge graphs enhance traditional KGs by assigning confidence scores to the facts or triples they contain, making it possible to represent uncertain information [5]. Advanced methods such as Markov Logic Networks [?] have been developed to propagate and infer uncertainty within the structure of a graph. Embedding techniques allow uncertain knowledge graphs

to be represented in a continuous vector space, supporting tasks like link prediction, entity classification, and fact verification [5]. These methods address challenges like missing, noisy, or conflicting data, enabling more robust reasoning and decision-making. Applications of these approaches include question answering systems and personalized recommendations, where the ability to handle uncertainty is crucial.

2.3 Uncertainty Quantification in LLMs

For Large Language Models (LLMs), uncertainty quantification (UQ) is particularly important as their outputs often lack clear indications of reliability. One widely used approach is calibration, which adjusts model confidence scores to better match actual accuracy [14]. Perturbation-based methods have also gained traction; these involve introducing small changes to the input to measure how stable the model's responses are [14]. Another approach is embedding-based analysis, which incorporates semantic checks to evaluate the consistency and reliability of generated text [25]. Tools like FACTSCORE go further by breaking down generated responses into individual facts, comparing these facts against a reference source, and evaluating factual accuracy [30]. UQ methods for LLMs are often categorized based on whether they require access to the model's internal mechanics. White-box methods, for instance, rely on logit-based evaluations that assess sentence uncertainty by analyzing token-level probabilities or entropy [22, 31, 46]. These approaches contrast with black-box methods, which operate independently of the model's internal structure. The importance of UQ in LLMs extends to high-stakes areas such as medical decision-making [19] and content moderation, where ensuring reliability and trustworthiness is critical. As LLMs are increasingly applied in diverse real-world scenarios, UQ remains an essential area of research for improving their interpretability and reliability.

3 Methods

Our method is a black-box method intended to estimate the uncertainty of LLMs' outputs based on knowledge graphs. It consists of three parts, and the framework is illustrated in Figure 1.

- (1) Generate responses from LLMs.
- (2) Construct a set of knowledge graphs for the responses.
- (3) Estimate uncertainty by computing similarities between knowledge graphs.

For a given query prompt q_a , let $\mathbf{R} = \{r_1, r_2, r_3, \dots\}$ denote all the sampling responses generated by a LLM. As discussed, determining the similarity between long text responses is inherently challenging due to the intricate semantic relationships embedded across sentences and paragraphs. To address this, we propose decomposing each long text response r_i into a collection of simple sentences that capture the core ideas of the text. This decomposition is achieved by constructing a set of knowledge graphs $\mathbf{G}$ for the responses, where each knowledge graph g_i comprises a set of triples $\text{triples}_i = \{\text{triple}_{i_1}, \text{triple}_{i_2}, \text{triple}_{i_3}, \dots\}$. Each triple consists of a head, a relationship, and a tail, analogous to the subject, predicate, and object in grammatical structure, collectively representing a specific semantic fact extracted from the original text. By assembling these triples, we can effectively distill the complex

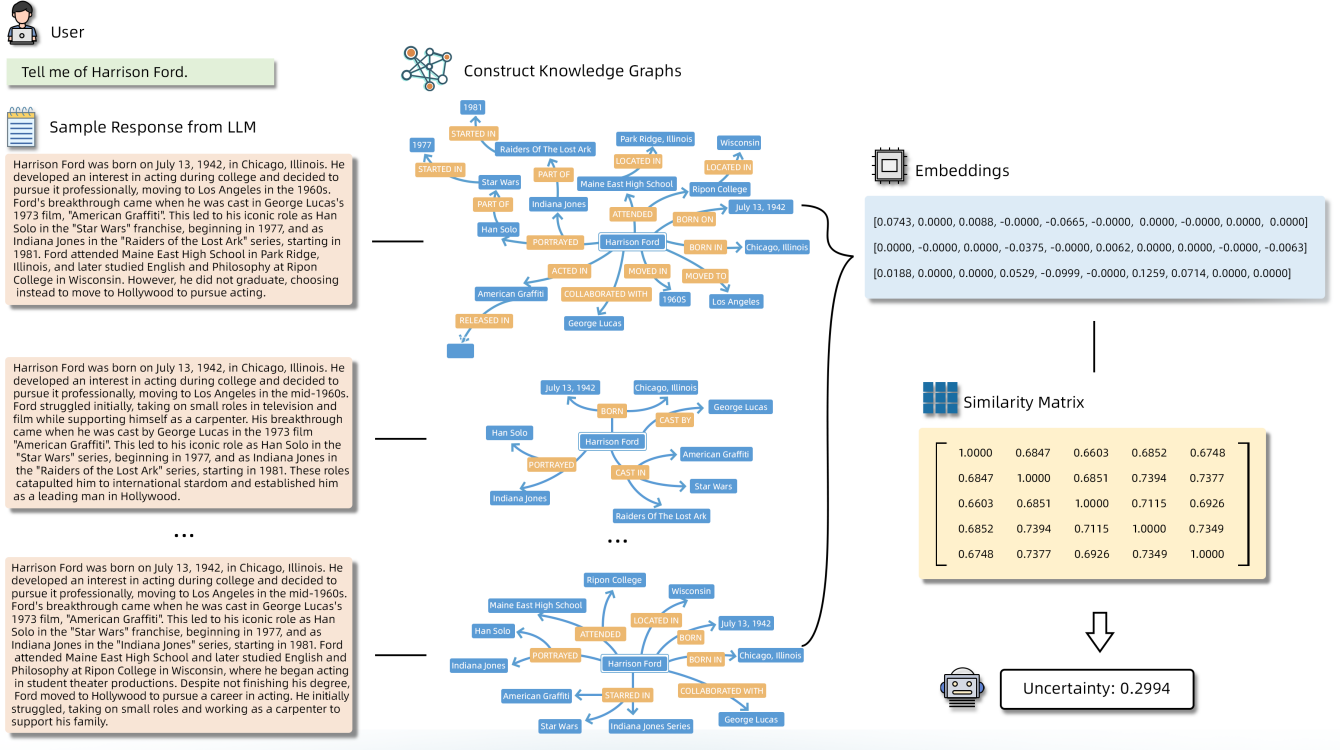


Figure 1: Overview of KG-UQ. For a given user query, we generate response n times using a LLM. Each response is then used to construct a corresponding knowledge graph. Embeddings are generated for each KG, and a similarity matrix is computed based on these embeddings. From this matrix, we derive the uncertainty associated with the LLM's responses to the user query.

semantic structure of the long text into a concise format.

$$r_i \xrightarrow{\text{Construct Knowledge Graph}} g_i = \{triple_{i_1}, triple_{i_2}, triple_{i_3}, \dots\}$$

$$triple_{i_j} = (head_{i_j}, relation_{i_j}, tail_{i_j})$$

To determine if facts $triple_{i_1} = \{triple_{i_1}, triple_{i_2}, triple_{i_3}, \dots\}$ that are decomposed from a long response are supported by others, we concatenate these triples into a simplified paragraph p_i and embed it into latent space. This approach ensures that the semantic content of the triples is preserved in a form suitable for embedding into a shared latent space. The process can be formalized as follows:

$$G = \{g_1, g_2, g_3, \dots\} \xrightarrow{\text{Concat Triples}} P = \{p_1, p_2, p_3, \dots\}$$

$$P = \{p_1, p_2, p_3, \dots\} \xrightarrow{\text{Embed}} E = \{e_1, e_2, e_3, \dots\}$$

where G represents the collection of knowledge graphs constructed from responses R , P represents the corresponding simplified paragraphs after concatenating the triples, and E represents the embeddings of these paragraphs in the latent space.

All responses generated using the same prompt are embedded into the same latent space, enabling semantic comparison between them. Specifically, the Euclidean distance in this space represents the semantic distance between responses. Let D denote the semantic distance matrix, where d_{ij} represents the semantic distance between paragraph p_1 and p_2 , corresponding to response r_i and r_j .

The semantic distance matrix is defined as:

$$D = \begin{pmatrix} d_{00} & d_{01} & \dots & d_{0n} \\ d_{10} & d_{11} & \dots & d_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n0} & d_{n1} & \dots & d_{nn} \end{pmatrix}$$

To quantify the confidence score of each response r_i concerning the query prompt q_a , we define the confidence score $C(r_i, q_a)$ as the average semantic distance between r_i and all other responses:

$$C(r_i, q_a) = \frac{1}{n} \sum_{\substack{j=0 \\ i \neq j}}^n d_{ij}$$

The overall uncertainty U for the query prompt q_a is then calculated as the average confidence score across all responses:

$$U(q_a) = \frac{1}{n} \sum_{i=0}^n C(r_i, q_a)$$

Another approach to obtaining the embedding of each knowledge graph is by leveraging a Graph Convolutional Network (GCN) [21]. In this framework, each knowledge graph is treated as a heterogeneous graph structure, where the head and tail of a triple are modeled as nodes in the graph, and the relationship between them serves as the edge connecting these nodes. A key aspect of this approach is the utilization of node types, which serve as an essential

property to distinguish the nodes. For example, consider a head node like *Donald Trump*. Depending on the context, this node could carry the property of being a *Person*, a *President*, or a *Real Estate Entrepreneur*. Node types allow the GCN to effectively encode such distinctions, enabling the model to better capture the semantic and structural nuances of the knowledge graph. The process can be formalized as follows:

$$G = \{g_1, g_2, g_3, \dots\} \xrightarrow{\text{Graph Embedding}} GE = \{ge_1, ge_2, ge_3, \dots\}$$

where GE denotes the set of graph embeddings generated by GCN.

Similarly, we can compute the pairwise distance matrix and uncertainty.

4 Experiments

4.1 Dataset and Evaluation Metric

We followed the experimental settings of LUQ [43] by utilising FACTSCORE [29], a fine-grained evaluation metric designed for factuality assessment of long-form text. FACTSCORE provides an automated method to estimate the factuality score of a given text by first decomposing it into atomic facts using LLMs and then verifying these facts against a custom knowledge source through retrieval. With a low error rate of just 2%, FACTSCORE ensures a high level of accuracy in quantifying the factuality of long text.

FACTSCORE also offers datasets for evaluation, which includes 183 names with human-annotated factuality labels corresponding to their respective Wikipedia titles, as well as an unlabeled dataset containing 500 additional names. To enhance the factuality assessment process, we batch-crawled page content from Wikipedia to construct our own comprehensive knowledge source. This custom knowledge source allowed us to further measure the factuality scores of text responses generated by various LLMs.

We determine the factuality score of a query prompt r_a by averaging the FACTSCOREs based on the responses. Then we use Pearson Correlation Coefficient (PCC) to quantify the linear relationship between the factuality scores and uncertainties. Additionally, we use the Spearman Correlation Coefficient (SCC) to measure the monotonic relationship between the two.

4.2 LLMs and Baseline Methods

We utilised six top-performing large language models (LLMs) to conduct our evaluation: ChatGPT-4o [35], ChatGPT-4 [1], ChatGPT-3.5-turbo [34], Llama-3.1-8B [9], Llama-3.1-70B [9], and Vicuna-33B [6]. This selection includes both lightweight models, such as Llama-3.1-8B, and larger parameter models, like Llama-3.1-70B and Vicuna-33B, along with the state-of-the-art ChatGPT series. These models provide a comprehensive range for assessing the capabilities of uncertainty quantification.

Our study follows the framework of LM-Polygraph [11], which implements a variety of uncertainty estimation methods. For white-box approaches, we selected three prominent methods: Maximum Sequence Probability (MSP), Monte Carlo Sequence Entropy (MCSE) [28], and Semantic Entropy (SE) [23]. These methods directly leverage the internal mechanics of LLMs for estimating uncertainty.

Additionally, we included several black-box methods as baselines, which operate independently of the internal states of the model:

Lexical Similarity (LexSim) [13], Number of Semantic Sets (NumSets) [24], Sum of Eigenvalues of the Graph Laplacian (EigV) [24], Degree Matrix (Deg) [24], Eccentricity (Ecc) [24], and Long-Text Uncertainty Quantification (LUQ) [43].

5 Uncertainty Quantification Results

Effectiveness for Open-Source Model Table 1 presents the correlation coefficients between the uncertainties derived from various uncertainty quantification methods and the factuality scores determined by FACTSCORE. Ideally, greater uncertainty in a LLM should correspond to lower factuality in its outputs. Our proposed methods demonstrate a strong correlation with factuality scores, particularly in open-source models such as Llama-3.1-8B, Llama-3.1-70B, and Vicuna-33B. The effectiveness of our method extends across both lightweight models with 8 billion parameters and larger-scale models with 70 billion parameters. While baseline approaches such as LUQ, LexSim, and EigV occasionally exhibit robustness, our methods consistently provide stability for these open-source models. NumSets interestingly displays a unique behavior, showing near-zero correlation, indicating its limited capability for uncertainty quantification in open-source models. Moreover, white-box methods consistently outperform black-box methods, delivering greater reliability and stability across different models.

For GPTs Uncertainty quantification methods, in general, exhibit low correlation between predicted uncertainties and factuality scores for GPTs models, highlighting limited effectiveness in accurately assessing uncertainty. Moreover, white-box methods are not applicable in this context due to the inaccessibility of internal model states. Certain methods, such as Deg and LUQ, even display positive correlations, which suggests poor uncertainty quantification performance, as higher uncertainty should ideally correspond to lower factuality. While our methods don't acquire the highest score for each model, they consistently maintain a strong correlation with factuality scores as measured by FACTSCORE, demonstrating the robustness and effectiveness in uncertainty quantification.

Effectiveness of Knowledge Graph As discussed, our method consists of three parts, with the construction of a knowledge graph serving as a preprocessing step to break down long texts into distinct statements, thereby simplifying their semantic structure. We compare the performance of the pipeline with and without the knowledge graph construction step, and Table 2 demonstrates its effectiveness. Knowledge graphs prove inherently beneficial for Llama models, significantly improving the correlation between uncertainty and FACTSCORE. However, the performance with knowledge graph took a dive with other models.

This disparity arises because constructing a knowledge graph from long text is inherently challenging, and current methods cannot guarantee capturing all key information points. Existing approaches, such as LangChain and REBEL[17], rely on LLMs[36], which exhibit inherent uncertainty. The information extracted from the same sentence may vary across iterations, and there is no mechanism to ensure that the triples generated by LLMs comprehensively cover all critical information points.

Additionally, the expectation that a knowledge graph should encapsulate all aspects of a long text may be inherently unrealistic.

Table 1: Performance Comparison of Various Methods with Different LLMs

		White-Box Methods			Black-Box Methods								
		MSP	MCSE	SE	LexSim	Ecc	NumSets	EigV	Deg	LUQ	Our	OpenAI	OpenAI-KG
FACTSCORE-BIO													
GPT-4o	PCC	-	-	-	-0.4954	0.0335	-0.1553	-0.2367	0.1095	0.1948	0.1253	-0.6647	-0.4577
	SCC	-	-	-	-0.4449	0.0967	-0.1404	-0.0041	0.2158	0.1790	-0.1970	-0.4634	-0.2588
GPT-4	PCC	-	-	-	0.0908	0.0918	-0.2399	-0.1534	0.1469	0.2385	-0.1922	-0.4207	-0.2395
	SCC	-	-	-	-0.3383	0.1184	-0.2415	-0.0967	0.1628	0.3445	-0.2552	-0.3849	-0.2433
GPT-3.5-turbo	PCC	-	-	-	-0.5520	-0.3636	-0.6334	-0.6817	-0.6201	0.6849	-0.1350	-0.6727	-0.5447
	SCC	-	-	-	-0.5083	-0.4389	-0.6058	-0.6441	-0.6403	0.6509	-0.1581	-0.5630	-0.4665
Llama-3.1-8B	PCC	-0.2634	-0.5035	-0.3933	0.1439	-0.2473	-0.0929	-0.5171	-0.3198	-0.6542	-0.3752	-0.2387	-0.5536
	SCC	-0.3550	-0.5088	-0.4991	-0.0109	-0.1906	-0.0936	-0.5640	-0.2955	-0.6492	-0.4777	-0.2566	-0.7978
Llama-3.1-70B	PCC	-0.3821	-0.6171	-0.5748	-0.2078	-0.3207	-0.2979	-0.6285	-0.4144	-0.3084	-0.3503	-0.6509	-0.6299
	SCC	-0.4306	-0.6259	-0.6082	-0.4185	-0.3197	-0.3072	-0.6802	-0.4172	-0.2697	-0.4780	-0.6204	-0.7303
Vicuna-33B	PCC	-0.5358	-0.7509	-0.7728	-0.7863	-0.2878	-0.1464	-0.5310	-0.4003	0.6150	-0.2183	-0.8355	-0.7715
	SCC	-0.5579	-0.7603	-0.7826	-0.7936	-0.1693	-0.1240	-0.5358	-0.3805	0.6520	-0.2323	-0.8317	-0.7735

Table 2: Performance Comparison of With and Without Knowledge Graph

Models	GPT-4o		GPT-4		GPT-3.5-turbo		Llama-3.1-8B		Llama-3.1-70B		Vicuna-33B	
	PCC	SCC	PCC	SCC	PCC	SCC	PCC	SCC	PCC	SCC	PCC	SCC
Without Knowledge Graph	-0.6647	-0.4634	-0.4207	-0.3849	-0.6727	-0.5630	-0.2387	-0.2566	-0.6509	-0.6204	-0.8355	-0.8317
With Knowledge Graph	-0.4577	-0.2588	-0.2395	-0.2433	-0.5447	-0.4665	-0.5536	-0.7978	-0.6299	-0.7303	-0.7715	-0.7735

For instance, consider the sentence from Botak Chin’s autobiography we got from LLM: *"His life took a drastic turn when he entered the world of crime at the age of 20, starting with petty thefts and gradually escalating to armed robberies."* This sentence comprises four distinct statements:

- His life took a drastic turn.
- He entered the world of crime at the age of 20.
- He started with petty thefts.
- He escalated to armed robberies.

Extracting all four statements simultaneously is challenging, and the extracted information often varies. For example, statement c could be rephrased as *"He began to steal petty things"*, which conveys the same semantic meaning but introduces significant differences in the resulting knowledge graph, thereby increasing uncertainty. Moreover, the extracted statements may fail to capture the full context of the original sentence. For instance, statement c alone does not convey that these events occurred when *"his life took a drastic turn."*

Although knowledge graphs are inherently effective at decomposing long text into discrete statements intended to capture the full semantic meaning of the original text, current methods fall short of achieving this goal. To our understanding, no approach to construct knowledge graphs that consistently extracts complete and identical information across iterations has been found.

6 Conclusion

In this work, we address the current limitations of uncertainty quantification methods for long text and explore the potential of knowledge graphs for their ability to transform unstructured text into structured data. This capability aids in deconstructing the complex semantic relationships inherent in long text. We propose

KG-UQ, a UQ method based on KGs, which decomposes long text into multiple statements that collectively preserve the semantic meaning of the original text. Instead of treating the text as a single entity or relying on sentence-level consistency, KG-UQ identifies the smallest semantic units within lengthy paragraphs, effectively overcoming the challenges associated with long text. Our findings demonstrate that KGs are an effective tool for improving the correlation between estimated uncertainties and factuality scores for both open-source and closed-source LLMs. However, the process of constructing KGs remains challenging due to its reliance on LLMs, which often results in capturing only partial semantic meaning from the original text. This work highlights the promising applications of knowledge graphs in UQ and lays the groundwork for future research to further refine and expand upon these methodologies.

7 Limitation

This work has several limitations, which present opportunities for further exploration and improvement:

- **Knowledge Graph Construction Methods:** In this study, we construct KGs using LangChain and REBEL, both of which are LLM-based methods. However, the output KGs generated by these methods often fail to fully capture the complete semantic meaning of long text. Traditional methods for KG construction, which may offer complementary strengths, were not investigated and should be explored in future work.
- **Evaluation Metrics:** We rely on FACTSCORE as the primary metric to evaluate the correlation between estimated uncertainties and factuality scores. While effective, FACTSCORE may not always reflect real-world factuality accurately. Incorporating human evaluation-based methods could provide

a more precise assessment of the factuality of generated text, offering valuable insights for model improvements.

- **Temperature Sensitivity:** We observe that temperature plays a crucial role in uncertainty quantification. Specifically, higher temperature settings often result in stronger correlations between uncertainties and factuality scores. Future studies could delve deeper into how temperature and other decoding parameters influence uncertainty estimation across different models and datasets.
- **Scalability and Efficiency:** While KGs are effective in handling the semantic complexity of long text, their construction process can be computationally expensive, especially for large-scale applications. Optimizing KG generation for scalability and efficiency would be a valuable direction for future research.

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